# 100 C++ Programming Problems

Aviral's Practice Sheet

Basic to Advanced · No STL · No DSA · No OOP

## Level 1 — I/O & Math

| | | |
|---|---|---|
| 01 | Print Hello World | [ ] |
| 02 | Read two integers, print their sum | [ ] |
| 03 | Average of two numbers | [ ] |
| 04 | Swap two numbers (without temp variable) | [ ] |
| 05 | Find largest of two numbers | [ ] |
| 06 | Find largest of three numbers | [ ] |
| 07 | Check even or odd | [ ] |
| 08 | Check positive, negative, or zero | [ ] |
| 09 | Simple calculator (+, -, *, /) using switch | [ ] |
| 10 | Area and perimeter of rectangle | [ ] |
| 11 | Area of circle given radius | [ ] |
| 12 | Celsius to Fahrenheit conversion | [ ] |
| 13 | Find quotient and remainder | [ ] |
| 14 | Check leap year | [ ] |
| 15 | Count digits in a number | [ ] |

## Level 2 — Loops

| | | |
|---|---|---|
| 16 | Print 1 to N | [ ] |
| 17 | Sum of first N natural numbers | [ ] |
| 18 | Factorial of N | [ ] |
| 19 | Fibonacci series up to N terms | [ ] |
| 20 | Check prime number | [ ] |
| 21 | Print all primes from 1 to N | [ ] |
| 22 | Check Armstrong number | [ ] |
| 23 | Sum of digits of a number | [ ] |
| 24 | Reverse a number | [ ] |
| 25 | Check palindrome number | [ ] |
| 26 | GCD of two numbers | [ ] |
| 27 | LCM of two numbers | [ ] |

| | | |
|---|---|---|
| 28 | Power of a number (x^n) without built-in | [ ] |
| 29 | Multiplication table of N | [ ] |
| 30 | Count even and odd numbers from 1 to N | [ ] |

## Level 3 — Patterns

| | | |
|---|---|---|
| 31 | Right triangle of stars (n rows) | [ ] |
| 32 | Inverted right triangle of stars | [ ] |
| 33 | Full pyramid of stars | [ ] |
| 34 | Inverted pyramid of stars | [ ] |
| 35 | Diamond pattern of stars | [ ] |
| 36 | Right triangle with numbers | [ ] |
| 37 | Floyd's triangle | [ ] |
| 38 | Pascal's triangle | [ ] |
| 39 | Hollow square pattern | [ ] |
| 40 | Butterfly pattern | [ ] |
| 41 | Number pyramid | [ ] |
| 42 | Alphabet triangle (A, AB, ABC...) | [ ] |

## Level 4 — Functions & Recursion

| | | |
|---|---|---|
| 43 | Function to check prime | [ ] |
| 44 | Function to return factorial | [ ] |
| 45 | Recursive factorial | [ ] |
| 46 | Recursive Fibonacci | [ ] |
| 47 | Recursive sum of digits | [ ] |
| 48 | Recursive reverse of number | [ ] |
| 49 | Recursive GCD (Euclidean algorithm) | [ ] |
| 50 | Recursive power (x^n) | [ ] |
| 51 | Recursive print 1 to N | [ ] |
| 52 | Function overloading — add int, add float, add string | [ ] |
| 53 | Default arguments in function | [ ] |
| 54 | Pass by value vs pass by reference demo | [ ] |
| 55 | Recursive count of zeros in a number | [ ] |

## Level 5 — Arrays

| | | |
|---|---|---|
| 56 | Find max and min in array | [ ] |
| 57 | Sum and average of array | [ ] |
| 58 | Reverse an array | [ ] |
| 59 | Check if array is sorted | [ ] |
| 60 | Linear search in array | [ ] |
| 61 | Binary search in sorted array | [ ] |
| 62 | Count occurrences of element in array | [ ] |
| 63 | Remove duplicates from sorted array | [ ] |
| 64 | Rotate array by K positions | [ ] |
| 65 | Merge two sorted arrays | [ ] |
| 66 | Bubble sort | [ ] |
| 67 | Selection sort | [ ] |
| 68 | Insertion sort | [ ] |
| 69 | Second largest element in array | [ ] |
| 70 | Matrix addition (2D arrays) | [ ] |

## Level 6 — Strings

| | | |
|---|---|---|
| 71 | Length of string without strlen | [ ] |
| 72 | Reverse a string | [ ] |
| 73 | Check palindrome string | [ ] |
| 74 | Count vowels and consonants | [ ] |
| 75 | Count words in a sentence | [ ] |
| 76 | Convert lowercase to uppercase manually | [ ] |
| 77 | Check if two strings are anagrams | [ ] |
| 78 | Remove spaces from string | [ ] |
| 79 | Find most frequent character in string | [ ] |
| 80 | Check if string is all digits | [ ] |
| 81 | Binary to decimal conversion | [ ] |
| 82 | Decimal to binary conversion | [ ] |
| 83 | String compression (aabbc -> a2b2c1) | [ ] |

## Level 7 — Pointers & Memory

| | | |
|---|---|---|
| 84 | Basic pointer — print address and value | [ ] |

| | | |
|---|---|---|
| 85 | Swap using pointers | [ ] |
| 86 | Pointer to array — traverse and print | [ ] |
| 87 | Dynamic array using new and delete | [ ] |
| 88 | Dynamic 2D array using new | [ ] |
| 89 | Function returning pointer | [ ] |
| 90 | Reference vs pointer demo | [ ] |
| 91 | Implement your own strlen using pointer | [ ] |

## Level 8 — Mixed Challenge

| | | |
|---|---|---|
| 92 | Check if number is perfect (sum of divisors = number) | [ ] |
| 93 | Print all Armstrong numbers between 1 and 1000 | [ ] |
| 94 | Find all factors of a number | [ ] |
| 95 | Roman numeral to integer | [ ] |
| 96 | Caesar cipher (encrypt a string by shifting) | [ ] |
| 97 | Matrix multiplication (2D arrays) | [ ] |
| 98 | Find duplicates in unsorted array | [ ] |
| 99 | Longest word in a sentence | [ ] |
| 100 | Number to words (1->'one', 21->'twenty one') | [ ] |

Complete all 100 before starting Striver's A2Z DSA sheet. · Platform: HackerRank C++ domain